Declaration of the Use of Generative AI for Assessment Work

Student name: _____

Student ID: _____

Module Code: _____

Assessment Title: _____

Please make sure you understand the policy about using generative AI, and such use must be explicitly acknowledged. The teacher/supervisor responsible for the assessment will provide specific guidance regarding acceptable usage, based on the module learning outcomes and marking criteria of the assessment. Failure to follow the rules for your work constitutes academic dishonesty.

Please tick either one below.

☐ I have used generative AI in this assessment work

☐ I did not use generative AI in this assessment work (please go to sign)

Generative AI tool used (*name and version*): _____

For each relevant area, briefly describe how the information was generated, as well as explain how the output was used in your work. Put N/A if not appropriate.

Type of Usage	Done by AI tools	Done by student
Brainstorming and idea generation		
Rephrasing text or formulas or generating text applied directly to the task		
Generating literature/background review		
Coding programming tasks		
Producing diagrams/figures		
Analysing data		
Other (please specify):		

Student signature:
Date:

CSAI2121

Probability and Statistics

Assignment 1

Student Information

Student ID: _____

Name: _____

Total Score _____ / 100

For grading use:

Total: 6 Main Questions, 100 Marks

1. Independence from a Joint Probability Table

The joint distribution of two discrete random variables X and Y is given by

	$Y=0$	$Y=1$	$Y=2$
$X=0$	0.10	0.15	0.05
$X=1$	0.20	0.30	0.20

Define the events

$$A=\{X=1\}, \qquad B=\{Y\geq 1\}, \qquad C=\{Y=0\}.$$

Total marks for this question: [15]

1.1 Find the marginal distribution of X and the marginal distribution of Y. **[4 marks]**

(Write your answer here.)

1.2 Find $P(A)$, $P(B)$, and $P(A \cap B)$. **[4 marks]**

(Write your answer here.)

1.3 Determine whether A and B are independent, and determine whether B and C are mutually exclusive. **[4 marks]**

(Write your answer here.)

1.4 Find $P(A \mid B)$ and $P(A \cup B)$. **[3 marks]**

(Write your answer here.)

2. Transforming a Confidence Score

A confidence score X has PDF

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

A transformed score is defined by

$$Y = 100X + 10.$$

Total marks for this question: [15]

2.1 Find $E[X]$ and $\mathrm{Var}(X)$, and use these results to find $E[Y]$ and $\mathrm{Var}(Y)$. **[6 marks]**

(Write your answer here.)

2.2 Find $P(Y > 80)$ and determine the range of possible values of Y. **[4 marks]**

(Write your answer here.)

2.3 Find the CDF $F_Y(y)$ and the PDF $f_Y(y)$. **[5 marks]**

(Write your answer here.)

3. Bayesian Decision-Making with Unequal Costs

A fraud-detection system examines online transactions. The probability that a transaction is fraudulent is

$$P(F) = 0.005.$$

The system raises an alert A with probabilities

$$P(A \mid F) = 0.98, \qquad P(A \mid F^c) = 0.02.$$

If an alerted transaction is blocked, the cost is \$20 when the transaction is legitimate and \$5 when the transaction is fraudulent. If an alerted transaction is allowed, the cost is \$1,000 when it is fraudulent and \$0 when it is legitimate.

Total marks for this question: [18]

3.1 Find $P(A)$ and $P(F \mid A)$. **[6 marks]**

(Write your answer here.)

3.2 Find the conditional expected cost of blocking an alerted transaction and the conditional expected cost of allowing an alerted transaction, and determine the cost-minimizing decision after an alert. **[8 marks]**

(Write your answer here.)

3.3 Let $p = P(F \mid A)$. Derive the threshold value of p above which blocking has a smaller expected cost than allowing. **[4 marks]**

(Write your answer here.)

4. Inferring a Density from Its Mean

A continuous random variable X has PDF

$$f_X(x) = \begin{cases} a + bx, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

It is known that

$$E[X] = \frac{7}{12}.$$

Total marks for this question: [15]

4.1 Use the normalization condition and the given mean to obtain two equations involving a and b, solve for a and b, and verify that the resulting function is nonnegative on $[0, 1]$. **[6 marks]**

(Write your answer here.)

4.2 Find the CDF of X. **[4 marks]**

(Write your answer here.)

4.3 Find the median of X, and find $E[X^2]$ and $\mathrm{Var}(X)$. **[5 marks]**

(Write your answer here.)

5. Designing a Two-Sided Normal Alarm Threshold

Under normal operating conditions, a sensor reading X follows

$$X \sim N(100, 5^2).$$

An alarm is raised when

$$|X - 100| > c.$$

Total marks for this question: [17]

5.1 Find c so that the false-alarm probability is 0.05, and express the alarm region as two numerical intervals. **[6 marks]**

(Write your answer here.)

5.2 Suppose a fault changes the distribution to $X \sim N(112, 5^2)$; find the probability that the alarm detects the fault. Repeat the calculation if the fault distribution is instead $X \sim N(105, 5^2)$. **[7 marks]**

(Write your answer here.)

5.3 Find c if the false-alarm probability is reduced to 0.01. **[4 marks]**

(Write your answer here.)

6. Repeated Testing for a Rare Condition

A condition occurs in 1% of the population. A diagnostic test has sensitivity 0.95 and false-positive probability 0.05:

$$P(+ \mid D) = 0.95, \qquad P(+ \mid D^c) = 0.05.$$

A person is tested three times. Assume that the test results are conditionally independent given the person's true condition.

An alarm is raised if at least two of the three tests are positive.

Total marks for this question: [20]

6.1 Find the probability that the alarm is raised given that the person has the condition, and the probability that the alarm is raised given that the person does not have the condition. **[5 marks]**

(Write your answer here.)

6.2 Find the overall probability that the alarm is raised, the probability that the person has the condition given that the alarm is raised, and the false-negative probability of the three-test alarm rule. **[9 marks]**

(Write your answer here.)

6.3 Given one positive result from a single test, find the probability that the person has the condition, and compare this posterior probability with the posterior probability obtained from the three-test alarm rule. **[6 marks]**

(Write your answer here.)

End of Assignment 1